

HW 4
30/10/2025

10 Points Possible

Due Oct. 30

4.1 # 7, 12, 17, 18, 19

4.2 # 1, 5

Homework 4 Problems (from 3rd edition textbook) 

4.1: Matrix Algebra

7) Let A have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are $n \times n$. Prove that if $B - I$ is nonsingular, then, for $k \geq 1$,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

$$A^1 = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}.$$

$$A^2 = \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

$$A^3 = \begin{bmatrix} B^3 & B^2C + BC + C \\ 0 & I \end{bmatrix}$$

$$A^k = \begin{bmatrix} B^k & (B^{k-1} + B^{k-2} + \dots + I)C \\ 0 & I \end{bmatrix}$$

$$\leadsto (\beta^2 - I)(\beta - I)^{-1}C = Bc + c?$$

$$S_0 = I$$

$$S_1 = B + I$$

$$S_2 = B^2 + B + I$$

$$S_n = B^n + B^{n-1} + \dots + B^1 + I$$

$$B S_{n-1} = B^n + B^{n-1} + \dots + B^1$$

$$(S_n - B S_{n-1}) = I$$

$$S_n = S_{n-1} + B^n$$

$$S_n = B S_{n-1} + I$$

$$B S_n = B S_{n-1} + B^{n+1}$$

$$(B - I) S_n = (B^{n+1} - I)$$

$\rightarrow$ of $B - I$ is NS, & L: